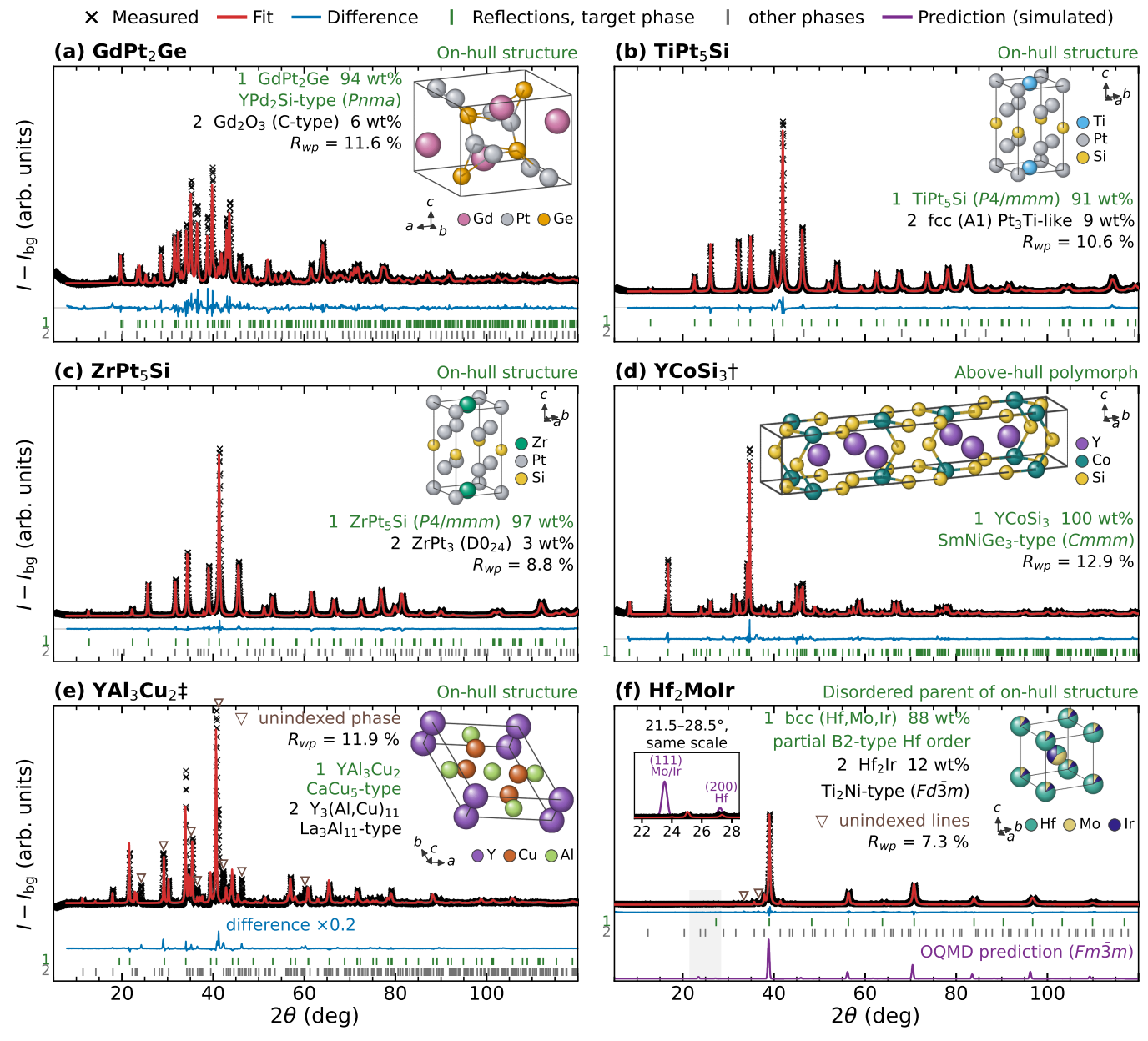

Measured
Fit
Difference
Reflections, target phase
other phases
Prediction (simulated)
(a) GdPt2Ge
On-hull structure
1 GdPt2Ge 94 wt%
YPd2Si-type (Pnma)
2 Gd2O3 (C-type) 6 wt%
Rwp = 11.6 %
Gd Pt Ge
(b) TiPt5Si
On-hull structure
Ti
Pt
Si
1 TiPt5Si (P4/mmm) 91 wt%
2 fcc (A1) Pt3Ti-like 9 wt%
Rwp = 10.6 %
(c) ZrPt5Si
On-hull structure
Zr
Pt
Si
1 ZrPt5Si (P4/mmm) 97 wt%
2 ZrPt3 (D024) 3 wt%
Rwp = 8.8 %
(d) YCoSi3†
Above-hull polymorph
Y
Co
Si
1 YCoSi3 100 wt%
SmNiGe3-type (Cmmm)
Rwp = 12.9 %
(e) YAl3Cu2‡
On-hull structure
unindexed phase
Rwp = 11.9 %
1 YAl3Cu2
CaCu5-type
2 Y3(Al,Cu)11
La3Al11-type
Y Cu Al
difference ×0.2
(f) Hf2MoIr
Disordered parent of on-hull structure
21.5–28.5°,
same scale
(111)
Mo/Ir
(200)
Hf
1 bcc (Hf,Mo,Ir) 88 wt%
partial B2-type Hf order
2 Hf2Ir 12 wt%
Ti2Ni-type (Fd3̄m)
unindexed lines
Rwp = 7.3 %
Hf Mo Ir
OQMD prediction (Fm3̄m)
I − Ibg (arb. units)
2θ (deg)